

Nayan Kumar

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Education

Indian Institute of Technology (IIT ISM) Dhanbad Expected May 2028
Integrated Master of Technology in Mathematics & Computing **GPA: 8.87/10**

Publications

GRIM Project Website | [arXiv Paper](#)
*Shailesh, Alok Raj, **Nayan Kumar**, Priya Shukla, Andrew Melnik, Michael Beetz, Gora Chand Nandi*
AAAI'26 (Oral)

Experience

Center for Intelligent Robotics, IIIT Allahabad  Mar 2025 – Sep 2025
Research Intern – Computer Vision : Supervised by Prof. G.C. Nandi & Prof. Andrew Melnik

- Designed and implemented a **multi-view perception system** that fuses **2–4 RGB views** into a unified **3D point cloud** for robotic grasp prediction.
- Integrated **DINOv2 (768-D) visual features** and evaluated **3+ learning baselines** on **TaskGrasp**, achieving up to **0.67 AP** with **5–8%** gains on unseen objects.
- Validated performance through **20+ real-world trials** on a **Kinova Gen3 Lite**, using failure analysis to guide iterative refinements.

Projects

Multimodal AI Technical Interviewer (Hybrid Architecture)  Jan 2026 – Feb 2026
Personal Project

- Built a **real-time multimodal inference system** combining LLM reasoning, neural TTS, and streaming I/O, sustaining $< 300\text{ ms}$ per-turn latency on an **NVIDIA Tesla GPU (32 GB VRAM)**.
- Deployed a **hybrid edge–cloud architecture** using **SSH reverse tunneling** and **WebSockets** for secure, stateful interactions.
- Modeled interview flow as a **graph-based state machine** (LangGraph), enabling adaptive follow-up questions with $< 500\text{ ms}$ end-to-end conversational latency.

Amazon ML Challenge 2025: Smart Product Pricing  Oct 2025 – Nov 2025
Amazon ML Challenge 2025

- Built a **multimodal regression pipeline** over structured attributes and product text, establishing a **50.05% SMAPE** baseline.
- Conducted **systematic feature ablation** to quantify modality contribution, revealing text representations as the dominant predictive signal.
- Guided by ablation results, optimized a **text-only LightGBM model** with **TF-IDF embeddings (20k features)**, reducing error to **46.03% SMAPE**.

Skills

Languages: Python, C, C++
Machine Learning: Supervised Learning, Feature Engineering, Model Evaluation, Error Analysis, Multimodal Learning
Computer Vision: CNNs, Vision Transformers, Multi-View Geometry, 3D Point Clouds
Frameworks & Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Linear Algebra, Probability & Statistics, Statistical Inference, GPU Computing, Introduction to Artificial Intelligence

Achievements

- Top 2% Team — Amazon ML Challenge 2025** Oct 2025
- 3rd Place — HackFest**, 36-hour hackathon at IIT (ISM) Dhanbad Feb 2025
- Top 3.3% Finish — CodeFest'25** (National ICPC-style contest) Mar 2025
- Competitive Programming — CodeChef (Rating: 1700) | Codeforces (Rating: 1560)** Present

Leadership & Engagement

- Member, RoboISM** (Robotics & AI Club, IIT (ISM) Dhanbad) – contributed to team-based robotics and AI projects.
- Event Coordinator, Robowars @ Concetto'24** – coordinated competitions and managed logistics for **10+ teams**.
- Senior Mentor, JH-SIC Hackathon (NVCTI)** – mentored teams during a **3-day hackathon**.